



The 10th International Competition of the Military Technical College (ICMTC 2026)



Lt. General Ibrahim Selim Award For Innovation

Details, Rules, and Format

*Student teams are invited to compete and demonstrate their vehicles at the Satellite
Challenge held at the Military Technical College in*

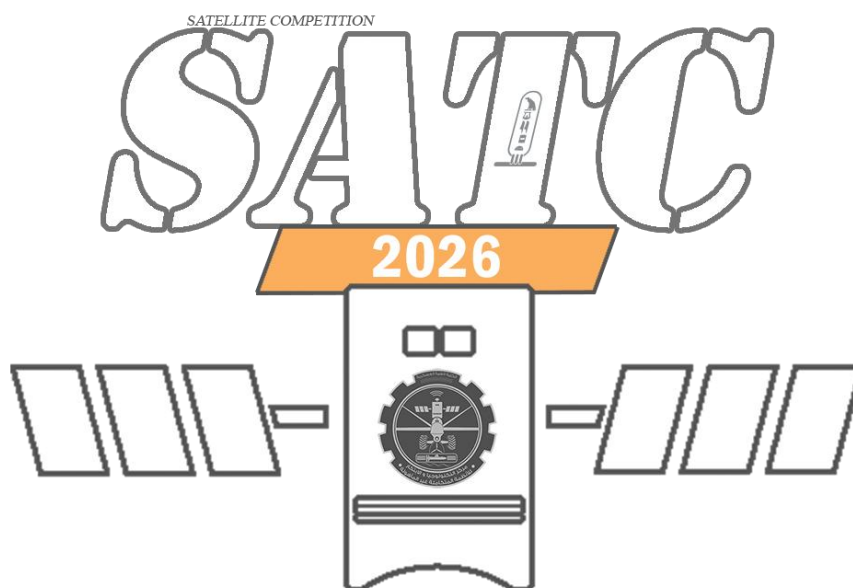
Kobry El-Kobba, Cairo, Egypt, on July 11th – July 16th, 2026



February 15th, 2026

The 10th International Competition of the Military Technical College

Satellite Challenge (SATC-6)





1. Introduction

1.1 Overview

The competition will engage university undergraduate students in the design, construction, development, and demonstration of satellite subsystems suitable for micro and small satellite applications. The competition awards, named in honor of Lt. General Ibrahim Selim, co-founder of the Military Technical College (MTC), aim to encourage students at all academic levels to participate in multidisciplinary teams and gain practical, hands-on experience in space system development.

Participants will operate within a structured team environment that mirrors real satellite missions, with clearly defined technical and non-technical roles. Engineering responsibilities may include subsystem design, sensor integration, embedded systems development, algorithm implementation, testing, and performance validation. In parallel, roles will be available for students from non-engineering disciplines, including project management, finance, documentation control, media coordination, outreach, and logistics.

The competition is designed to simulate the full engineering lifecycle of satellite subsystem development, fostering systems thinking, interdisciplinary collaboration, and professional-level execution consistent with aerospace industry practices.

1.2 Aims of the Competition

1. Providing participants with the opportunity to learn practical aerospace engineering skills and techniques for industry.
2. Providing the opportunity for space workers and students to experience practical activities.
3. Provide systems engineering participants with a challenge for a complex system that requires them to follow the industry-recognized engineering development life cycle (design, development and demo) against challenging mission requirements.
4. Provide an opportunity for participants to develop and demonstrate teamwork, leadership and business skills as well as technical proficiency.
5. Promoting job opportunities in the space sector, enhancing cooperation between universities in the field of space technology, and providing a forum for interdisciplinary research.
6. Providing students and specialists with the opportunity for learning, creativity, excellency and innovation.
7. Spreading the culture of space science, engineering and technology and their applications to a large segment of the public



2. General Rules

1. The competition is open to undergraduate students which can be affiliated to universities, institutes and research centers.
2. Teams may form an alliance to participate as a joint team.
3. Every team will submit an identified name at the beginning of the competition. The introduced team's name cannot be changed.
4. Every participating team is required to include three main members introduced at the beginning of the competition. The total number of participants in each team can reach up to 8 members, including students and supervisors.
5. The team leader is required to be one of the main members. The team leader will be the contact person of his corresponding team and all the communication with the competition committee will be performed through the team leader.
6. The team leader can be changed once during the competition upon the consent of the majority of the officially three introduced members.
7. Every participant can take part in only one team.
8. The unique official website of the SATC competition will be (<https://satc.conferences.ekb.eg>). Any updates for the rules will be announced in the official website as well as being sent to the team leaders.

3. Satellite Challenge tasks

The satellite challenge consists of **four tasks** as following:

Task I	Engineering Model Star Tracker
Task II	Engineering Model Sun Sensor
Task III	Smart Power System for Small Satellite
Task IV	Mission-Grade TT&C Communication System for CubeSats

All participating teams must complete the requirements of any selected two tasks. Additional bonus points will be awarded for completing the requirements of any additional tasks.



4. Schedule

Key dates and activities are as follows:

No.	Activity	Date
1	Competition Rules	15 st February 2026
2	Submission of Team's Proposal and Entry Forms	15 th March 2026
3	All Required Documents Submission Deadline	4 th June 2026
4	Acceptance from the Technical Committee	20 th June 2026
5	Demonstration Event	11 th July – 16 th July 2026

* Tentative dates: noting that adherence to deadlines is a prerequisite for next stage entry, and the organizer retains the right to eliminate any team in the event who will not submit all required deliverables on time.

5. Registration

1. The official source for all information concerning rules, interpretations, and information updates for SATC is the World Wide Web home page at:
<http://www.mtc.edu.eg/mtcwebsite/Competition.aspx>
2. Teams' registrations will be online via the following URL:
<https://satc.conferences.ekb.eg>
3. Teams' entry forms also will be submitted online no later than March 14, 2025.
4. An entry fee of **7000 EGP** for residents (**700 USD for internationals**) per team (maximum 10 members) is payable upon submission of an entry form (Registration). This fee contributes towards the cost of putting on the Demonstration Event. It is non-refundable if a team cannot participate in the Demonstration Event.
5. Entry fee payment date and options will be posted online or emailed to the successfully registered teams later.
6. An extra fee of **700 EGP** for residents (**70 USD for internationals**) per person for any extra team members and up to 5 members is payable on registration day (first day in the competition).
7. Any payments will be refunded **ONLY** in case of rejected teams.
8. The team proposal form must be submitted online through the SATC website no later than March 15, 2026. The main team supervisor must sign the team proposal form and certify that all student team members (Under graduates) are currently registered in the **academic year 2025/2026**.
9. Registered teams will submit their future on a shared google drive created by the competition board. The competition board will provide each registered team with their Google Drive shared link.



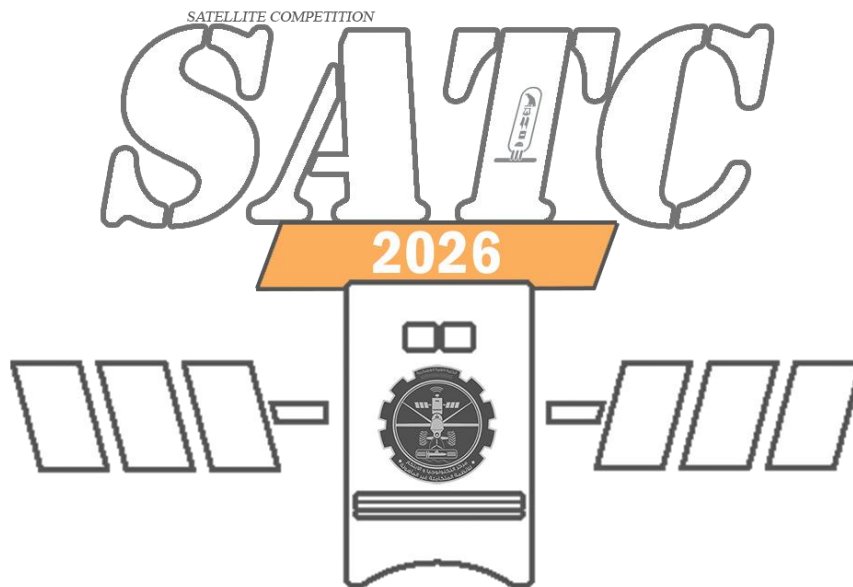
10. Teams' deliverables via shared google drive include the following team information that must be provided after registration immediately:

- Name of the team (and name of the vehicle in case it's different).
- Team members' full names.
- Photocopy of the national ID (for Egyptian team members).
- Photocopy of the passport (for non-Egyptian team members).
- Personal cell phone no., email, and address.
- List of sponsors.



The 10th International Competition of the Military Technical College

Satellite Challenge (SATC-6)



Task I: Engineering Model Star Tracker



1. Introduction

This document defines the technical and functional requirements for the design, development, and validation of an Engineering Model (EM) Star Tracker intended for use in a small satellite platform.

The project is proposed as an undergraduate engineering competition aimed at engaging students in real-world spacecraft subsystem design.

The Engineering Model Star Tracker shall:

- Be implemented using Commercial Off-The-Shelf (COTS) components.
- Demonstrate full star tracker functionality including:
 - Image acquisition
 - Image processing
 - Star centroid extraction
 - Pattern recognition
 - Star catalog identification
 - Autonomous attitude determination
- Operate without reliance on space-qualified or radiation-hardened components.
- Function as a fully integrated, self-contained subsystem.

2. Project Objectives

The main objectives of this competition are:

1. Design a flight-representative Engineering Model Star Tracker.
2. Implement real-time star image processing algorithms.
3. Perform autonomous attitude determination.
4. Data transfer using one of communication protocols.
5. Expose students to space-grade attitude determination engineering practices.

3. System Overview

The Engineering Model Star Tracker (EM-ST) shall:

- Capture star field images using an optical imaging sensor.
- Detect and centroid stars in the image.
- Perform pattern matching against an onboard star catalog.
- Determine spacecraft attitude quaternion.
- Estimate angular rates.
- Communicate telemetry via UART protocol.
- Operate in both:
 - Lost-in-space acquisition mode
 - Tracking mode



4. Functional Requirements

4.1. Attitude Determination Performance

The Engineering model shall determine:

- attitude quaternion (q_0, q_1, q_2, q_3)
- angular rate vector ($\omega_x, \omega_y, \omega_z$)

Performance requirements:

- $\leq 1^\circ$ cross-boresight (RMS) accuracy @ $3^\circ/\text{second}$
- $\leq 3^\circ$ around boresight (RMS) accuracy @ $3^\circ/\text{second}$

4.2. Update rate requirements

- The Engineering Model shall support:

Parameter	Requirement
Minimum update rate (tracking mode)	$\geq 1 \text{ Hz}$
Lost-in-space acquisition time	$< 2 \text{ seconds}$
Time stamping	Synchronized to internal reference clock

4.3. Communication Interface

The Engineering Model shall implement :

Feature	Requirement
Protocol	UART
Logic level	3.3 V
Bit rate	115200 bps

Each transmitted data frame shall include the following payload format as minimum:

- Quaternion components: q_0, q_1, q_2, q_3
- Angular rate vector: $\omega_x, \omega_y, \omega_z$
- Sensor reference clock (timestamp)
- Sensor status word (solution validity, tracking state, error flags)
- The detailed frame structure (header, payload format, checksum/CRC) shall be defined by the organizing committee prior to final testing.



4.4. Power Input Interface

The Engineering Model shall support:

Parameter	Requirement
Input voltage range	3.3 V – 12.0 V DC

The subsystem shall include internal onboard voltage regulation as required.

4.5. Power Consumption

Parameter	Requirement
Average power (tracking mode)	≤ 2.0 W

Power shall be measured at the DC input terminals.

5. Algorithm Requirements

5.1. Star Detection and Centroiding

The team shall describe:

- Star detection method
- Thresholding technique
- Centroid computation algorithm

5.2. Pattern Recognition

The team shall define:

- Pattern matching technique
- Lost-in-space identification method

5.3. Attitude Determination Algorithm

The documentation shall describe:

- Mathematical method used for attitude solution
- Angular rate estimation technique
- Filtering approach (if applicable)



6. Documentation Requirements

Each participating team shall submit a complete technical documentation package including:

6.1. System Architecture Document

- Block diagram
- Interface definitions
- Functional description

6.2. Algorithm Description

- Star detection and centroiding method
- Pattern recognition approach
- Catalog structure
- Attitude determination algorithm

7. Testing and evaluation

The evaluation and testing of the Engineering Model Star Tracker shall be implemented using the Static and Dynamic Star Tracker Simulator available at the AIT facility in the Space Technology Center (STC).

7.1. The static Star sensor

A static star sensor test is essential for validating the fundamental performance of a star tracker under controlled conditions before introducing dynamic motion effects. In static mode, the sensor observes a fixed star field, allowing accurate evaluation of centroiding precision, pattern recognition reliability, catalog matching accuracy, and absolute attitude determination performance without rotational disturbances. This test isolates algorithmic and optical errors from motion-induced errors, enabling verification of baseline RMS attitude accuracy, noise sensitivity, false identification rates, and calibration quality.

- **Passing the static test is mandatory to go through the competition evaluation.**



Figure 1 Static star sensor

7.2. The Dynamic Star sensor

Dynamic star sensor simulator is a component of AOCS test equipment, which is used to test the function and performance of star sensor in satellite control subsystem. It gives a star map which is replaced by the dynamic target simulation module in real time. The star map contains the exact star point position and the exact diagonal star spacing.

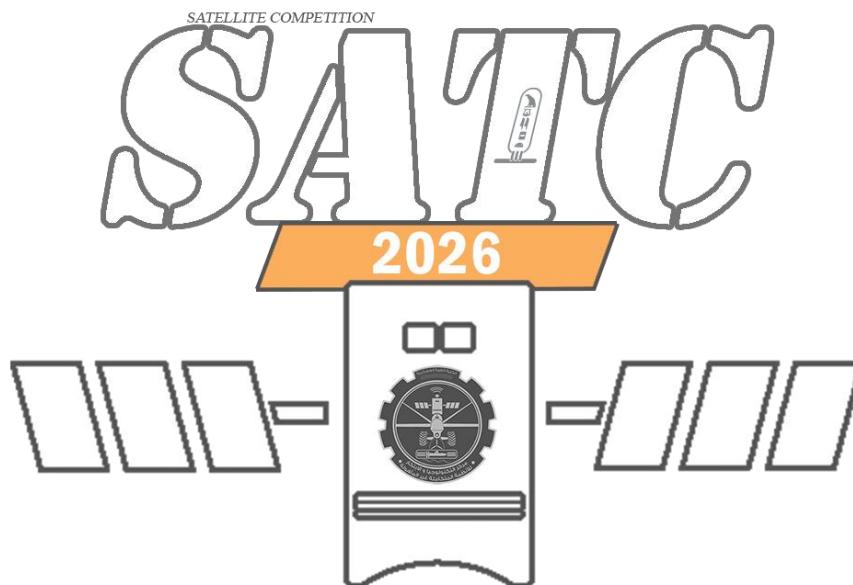


Figure 2 Dynamic star sensor



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Task II: Engineering Model Sun Sensor



1. Introduction

This document defines the technical and functional requirements for the design, development, and validation of a Coarse Sun Sensor (CSS) intended for use in a small satellite platform.

The project is proposed as a national undergraduate engineering competition, aimed at engaging students in real-world spacecraft subsystem design.

The CSS shall act as an attitude determination sensor used for determining the direction of the sun.

2. Project Objectives

The main objectives of this competition are:

1. Design a flight-representative Coarse Sun Sensor.
2. Implement a coarse sun sensor.
3. Use UART as the data interface.
4. Validate the design using hardware testing and simulation.
5. Expose students to space-grade engineering practices.

3. System Overview

The Coarse Sun Sensor (CSS) shall:

- Detect the sun illumination.
- Determine the sun direction.
- Communicate telemetry and receive commands via UART protocol.

4. Functional Requirements

4.1. Power Input Interface

- **Voltage Input Range**
 - Operating range: **3.3 V – 5.5 V DC**

4.2. Power Consumption

The CSS shall support:

Parameter	Requirement
Maximum total power	< 0.5 W



5. Data Interface Requirements

5.1. UART Communication Protocol

The CSS shall implement:

Feature	Requirement
Protocol	UART
Bit rate	115200 bps
Mode	Telemetry & Telecommand
Node ID	Configurable

5.2. Telemetry Parameters

Minimum required telemetry:

- Sun incidence angle
- Sun presence indicator (sun detected / missing).
- Sensor temperature (Celsius).
- Sensor status word (validity flags, saturation warning, error flags)
- Sensor reference clock (timestamp)

6. Accuracy

The CSS shall achieve accuracy:

- $\leq 3^\circ$ RMS angular accuracy (within $\pm 60^\circ$ field of view).
- $\leq 5^\circ$ RMS angular accuracy (within $\pm 75^\circ$ field of view).

7. Field of View

Minimum Field of View: $\pm 60^\circ$ (cross-axis)

Extended performance up to $\pm 75^\circ$ is desirable.

Bonus scoring may be given for: Field of View $\geq \pm 80^\circ$

8. Update Rate

CSS shall provide angular measurements at a minimum rate of 5 Hz.

Response time to illumination change shall be ≤ 200 ms.

Bonus scoring may be given for: Update rate ≥ 10 Hz.



9. Physical and Environmental Characteristics

mass	≤ 50 grams
Dimensions	$\leq 65 \text{ mm} \times 50 \text{ mm} \times 30 \text{ mm}$
Mounting	2x M3 screws.
Temperature Range	0°C to +50°C

10. Documentation

Each participating team shall submit a complete technical documentation package including:

5.3. System Architecture Document

1. Block diagram of entire system
2. Optical/mechanical interface definitions
3. Electrical interface specifications
4. Functional description and theory of operation
5. Component selection rationale

5.4. Detector and Optics Description

1. Photodetector type and characteristics (photodiode, phototransistor, solar cell, etc.)
2. Optical configuration (single detector, quad-cell, linear array, coded mask, etc.)
3. Optical filtering design (if applicable)

5.5. Algorithm Description

1. Sun direction determination method
2. Temperature compensation algorithm
3. Sun presence detection logic
4. Error handling and fault detection

5.6. Calibration and Test Report

1. Calibration procedure description
2. Test setup photographs and diagrams
3. Measured accuracy across FOV
4. Linearity and temperature dependence data
5. Repeatability and stability results

11. Testing and evaluation

The evaluation and testing of the coarse sun sensors could be implemented using the solar simulator existing in the AIT facility in Space technology Center (STC). Testing the coarse sun sensor with the solar simulator is essential for validating the fundamental performance of the sun sensor under controlled conditions. The solar simulator will be used to illuminate the sun sensor from a certain and known directions w.r.t the sensor and check the detected direction of the incident light to accurately evaluate the performance of the sun sensors.



Figure 3 Solar Simulator 1

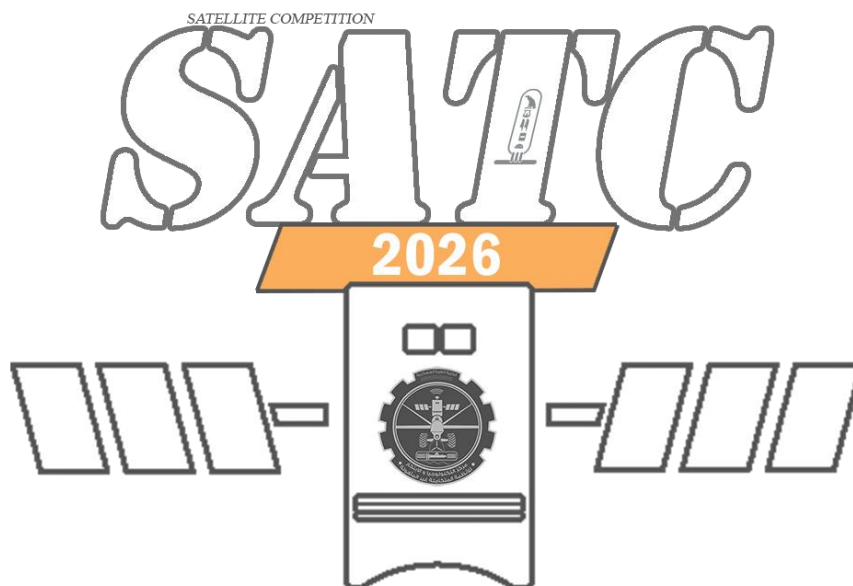


Figure 4 Solar Simulator 2



The 10th International Competition of the Military Technical College

Satellite Challenge (SATC-6)



Task III: Smart Power System for Small Satellite



1. Introduction

This document defines the technical and functional requirements for the design, development, and validation of a Power Control Module (PCM) intended for use in a small satellite platform.

The project is proposed as a national undergraduate engineering competition, aimed at engaging students in real-world spacecraft subsystem design.

The PCM shall act as the central unit responsible for power conditioning, distribution, monitoring, and protection for spacecraft subsystems.

2. Project Objectives

The main objectives of this competition are:

1. Design a flight-representative Power Control Module.
2. Implement power distribution and protection logic.
3. Use CAN Bus as the spacecraft data interface.
4. Validate the design using hardware testing and simulation.
5. Expose students to space-grade engineering practices.

3. System Overview

The Power Control Module (PCM) shall:

- Receive power from a satellite Electrical Power System (EPS).
- Distribute regulated power to multiple spacecraft loads.
- Monitor voltage, current, and temperature.
- Communicate telemetry and receive commands via CAN Bus.
- Provide electrical protection against faults.

4. Functional Requirements

4.1. Power Input Interface

The PCM shall accept electrical power through:

- **Umbilical Cable Interface**
 - Connector Model: **YF5-127Z**
 - Purpose: Ground testing, charging, and safe power injection.
- **Voltage Input Range**
 - Nominal input: **28 V DC**
 - Operating range: **20 V – 28 V DC**



4.2. Power Output Capability

The PCM shall support:

Parameter	Requirement
Maximum total power	200 W
Output channels	Minimum 4 independent channels
Channel control	Individually switchable
Output type	DC regulated
Efficiency	$\geq 85\%$

4.3. Load Management

The PCM shall:

- Enable or disable loads via software commands.
- Support **priority-based load shedding**.
- Provide soft-start for high inrush loads.

5. Data Interface Requirements

5.1. CAN Bus Interface

The PCM shall implement:

Feature	Requirement
Protocol	CAN 2.0B
Bit rate	125 kbps – 1 Mbps
Mode	Telemetry & Telecommand
Node ID	Configurable

5.2. Telemetry Parameters

Minimum required telemetry:

- Input voltage
- Input current
- Output voltage per channel
- Output current per channel
- Board temperature
- Fault flags



6. Protection and Safety Requirements

The PCM shall include:

5.3. Electrical Protections

Protection	Description
Overcurrent	Per channel electronic fuse
Overvoltage	Input transient protection
Undervoltage	Brown-out detection
Short-circuit	Automatic isolation
Reverse polarity	Input protection diode/MOSFET

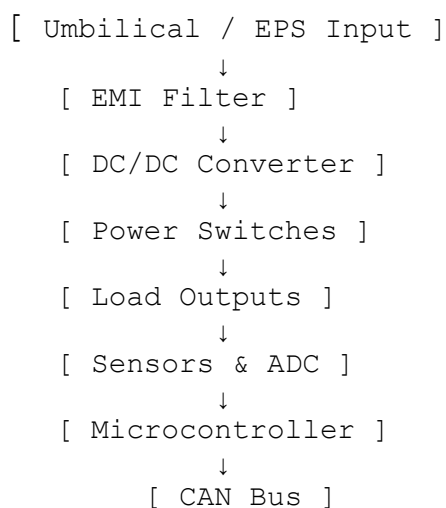
5.4. Fault Handling Logic

The PCM shall:

1. Detect fault condition.
2. Isolate affected channel.
3. Log fault via CAN.
4. Allow software reset.

7. Hardware Architecture

7.1. Main Blocks





7.2. Suggested Components

Function	Suggested Devices
MCU	STM32 / PIC32
CAN	MCP2562 / SN65HVD
Power switch	BTS500xx / TPS1H
DC/DC	LM25116 / Vicor
Current sensor	INA219 / ACS712

8. Software Requirements

8.1. Firmware Functions

The PCM firmware shall:

- Initialize hardware.
- Monitor sensors.
- Process CAN commands.
- Control power switches.
- Handle fault logic.
- Store system states.

8.2. Software Architecture

```
Main Loop
├── Sensor Task
├── CAN Task
├── Power Control Task
├── Fault Management Task
└── Housekeeping Task
```

9. Mechanical Requirements

Parameter	Requirement
Form factor	CubeSat-compatible
PCB type	4-layer
Mounting	M3 space-grade
Cooling	Passive thermal



10. Testing and Validation Plan

10.1. Electrical Tests

Test	Objective
Input voltage sweep	Verify regulation
Load test	Verify 200 W
Short-circuit test	Protection
CAN test	Communication

10.2. Environmental Tests (Bonus)

Test	Description
Thermal	-20°C to +60°C
Vibration	10 – 2000 Hz
EMI/EMC	CISPR 22

11. Deliverables

Each team must submit:

11.1. Technical Documents

1. System Architecture Document (SAD)
2. Electrical Schematics
3. PCB Layout
4. Bill of Materials (BOM)
5. Software Design Document (SDD)
6. Test Report

11.2. Hardware Deliverables

- One functional PCM prototype.
- CAN interface demo.
- Load test demo up to 200 W.

12. Evaluation Criteria

Criterion	Weight
Technical correctness	30%
Power performance	20%
Protection & safety	20%
Documentation	15%



Innovation	10%
Presentation	5%



Figure 5 Solar Array Simulator (SAS) (E4368A)



Figure 6 DC Power Supply (N8941A)



Figure 7 PXI Testing Equipment



Figure 8 DC Electronic Load (N3300A)

13. Competition Learning Outcomes

Students will gain experience in:

- Spacecraft power systems
- Embedded systems design
- CAN Bus protocol
- Power electronics
- Fault-tolerant systems
- Systems engineering workflow

14. Compliance with Space Standards (Reference)

Teams are encouraged to follow:

Standard	Purpose
ECSS-E-ST-20	Space system engineering



ECSS-Q-ST-30	Electronics
ECSS-Q-ST-60	Components
NASA GEVS	Environmental tests

15. Future Extensions (Advanced Category)

Optional advanced features:

- Redundant power buses.
- MPPT integration.
- Battery charge control.
- Telemetry encryption.
- AI-based fault detection.

16. Conclusion

This competition provides a realistic spacecraft subsystem engineering challenge, bridging academic learning with industrial space practices.

The Power Control Module designed under this specification can serve as a flight-representative prototype for CubeSat and small satellite missions.

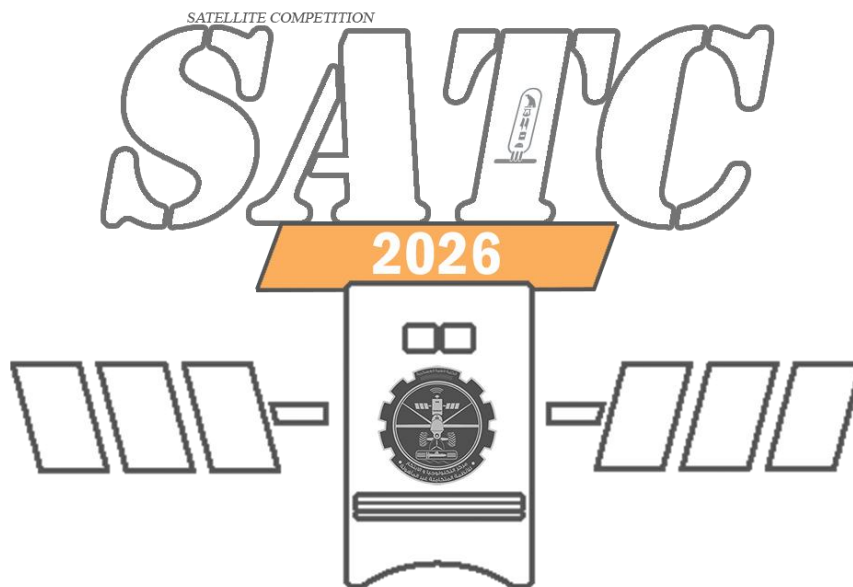
Document Classification

Field	Value
Document Type	Competition Technical Reference
Target Audience	Undergraduate Engineering Students
Level	Advanced
Domain	Space Systems Engineering
Validity	2026–2028



The 10th International Competition of the Military Technical College

Satellite Challenge (SATC-6)



Task IV: Mission-Grade TT&C Communication System for CubeSats



1. Introduction

CubeSats have evolved from educational platforms into operational spacecraft supporting scientific, commercial, and technology demonstration missions. As mission capabilities expand, the communication subsystem becomes a mission-critical element responsible for spacecraft control, safety, and operational continuity.

In low Earth orbit (LEO), communication links are subject to significant impairments including Doppler shift, additive noise, signal fading, and short visibility windows. Under these constraints, a spacecraft communication subsystem shall provide not only reliable signal transmission, but also robust synchronization, data integrity protection, controlled command execution, and validated closed-loop operation.

This competition call invites teams to design and verify a structured, engineering-grade telemetry, tracking, and command (TT&C) communication subsystem. The challenge reflects architectural principles used in real satellite missions while remaining achievable within an academic framework.

2. Project Objectives

The objective of this competition is to design, implement, and experimentally validate a fully functional TT&C communication subsystem suitable for CubeSat missions operating in LEO. The project emphasizes complete system integration, operational reliability, and quantitative performance evaluation under realistic orbital constraints.

Participating teams are required to develop a bidirectional communication subsystem capable of supporting telecommand uplink and telemetry downlink within short LEO visibility windows. The system shall implement a structured communication protocol including packet framing, error detection, and forward error correction to ensure reliable data exchange.

To ensure reliable mission operation, the subsystem shall include a defined reliability mechanism based on acknowledgment handling and controlled retransmission logic. In addition, the telecommand function shall incorporate basic security measures, including lightweight authentication and replay protection, to prevent unintended or unauthorized command execution.

A complementary ground segment will be designed to enable Doppler compensation, telecommand generation, telemetry decoding, and closed-loop operational verification. The complete communication chain shall be validated



experimentally using hardware testing, and system performance must be quantitatively assessed using defined key performance indicators (KPIs).

The final deliverable of each team will be an operational engineering model demonstrated through closed-loop testing and documented according to structured systems engineering practices.

3. System Overview

The communication system shall be structured as an integrated end-to-end architecture consisting of two coordinated elements: the onboard communication subsystem (Space Segment) and the supporting ground communication subsystem (Ground Segment). Together, these elements form a complete operational chain enabling command transmission, telemetry reception, and system-level interaction. The following subsections define the architectural boundaries, implementation flexibility, and structural expectations of both segments.

3.1 Space Segment Architecture

The space segment represents the onboard communication subsystem of the CubeSat. It shall include the radio-frequency interface, baseband processing capability, protocol handling functions, and system interfaces required to exchange data with other onboard subsystems. At a minimum, the onboard architecture shall contain:

- An RF transceiver or radio front-end
- A baseband processing unit (microcontroller or FPGA)
- Packet processing and protocol logic
- Interfaces to a simulated onboard data source and command execution unit

Teams are permitted to select between two implementation approaches:

- Custom baseband architecture: Teams may use discrete RF components or a transceiver integrated circuit while implementing packet formatting, coding, retransmission control, and command processing within their own embedded firmware or programmable logic.
- Hybrid architecture: Teams may use a commercial RF module for signal transmission and reception; however, higher-layer protocol handling, reliability control, and command management must be implemented independently. Fully integrated black-box communication solutions that conceal protocol behavior are not permitted.



The onboard communication subsystem shall comply with typical CubeSat resource constraints, particularly with respect to power budget, processing resources, and hardware integration.

3.2 Ground Segment Architecture

The ground segment serves as the operational interface between the user and the spacecraft communication subsystem. It shall provide the necessary hardware and software to generate telecommands, receive telemetry, and manage communication sessions.

Implementation may be based on Software Defined Radio (SDR) platforms or equivalent configurable radio architectures. The ground system shall include:

- A transmit chain for telecommand generation and modulation
- A receive chain for telemetry demodulation and decoding
- A user interface for command entry and telemetry visualization
- Data logging and session recording capabilities

The ground system shall allow controlled adjustment of communication parameters, such as frequency, modulation settings, data rate, and coding configuration, and shall support repeatable testing and quantitative performance evaluation.

3.3 Interface Definition and Data Flow

The communication system shall define clear internal and external interfaces. On the space segment, the communication subsystem must interface with a simulated onboard control unit capable of generating telemetry data and executing received commands. On the ground segment, operator interaction shall occur through a defined command interface and telemetry display environment.

Data flow shall be structured and traceable from command generation to onboard execution and from telemetry source to ground reception. Packet formats, headers, and state transitions must be explicitly defined to ensure transparent system behavior.

4. Functional Requirements

This section defines the mandatory functional capabilities of the communication subsystem. The requirements apply to both the space and ground segments and establish the minimum technical features necessary to achieve a complete and operational TT&C implementation.



4.1 Physical Layer Requirements

The physical layer shall ensure reliable signal transmission and reception under simulated LEO operating conditions. The implementation shall include the essential signal processing functions required for packet detection, synchronization, and error control. At a minimum, the implementation shall incorporate:

- A preamble structure enabling reliable packet detection
- Symbol timing recovery mechanisms
- Carrier frequency and phase recovery
- Doppler frequency tracking or compensation capability
- Forward Error Correction (FEC)
- Cyclic Redundancy Check (CRC) for frame integrity
- Interleaving techniques to mitigate burst errors

Teams shall justify the selection of modulation scheme, coding method, and data rate through a documented link budget analysis and robustness evaluation.

4.2 Link Layer Requirements

The link layer shall provide structured data exchange and support reliable telecommand and telemetry transfer. It shall ensure correct packet organization, delivery control, and error handling during communication sessions.

The link-layer implementation shall include:

- Defined packet framing structure
- Header fields and unique identifiers
- Sequence numbering for packet tracking
- Frame integrity verification
- A retransmission mechanism based on either stop-and-wait or selective repeat strategy
- Clearly defined timeout and retry policies

The retransmission design shall be consistent with short communication windows and must demonstrate appropriate handling of packet loss and acknowledgment management.

4.3 Command Protection Requirements

The telecommand channel shall include basic safeguards to ensure that only valid and authorized commands are executed by the spacecraft. The system shall implement:

- Lightweight command authentication (e.g., short message authentication code)



- Replay protection using sequence counters, frame counters, or timestamps

The onboard subsystem must be capable of detecting and rejecting unauthorized, corrupted, or previously used commands.

4.4 Ground Segment Functional Requirements

The ground segment shall provide the operational capabilities necessary to transmit telecommands, receive telemetry, and monitor communication performance. The ground system shall support:

- Telecommand generation and uplink transmission
- Telemetry reception and decoding
- Doppler correction or compensation capability
- Automated pass management, including acquisition and loss-of-signal handling
- Data logging and telemetry archiving
- Real-time monitoring of link performance indicators

The ground implementation shall reflect structured operational procedures rather than simple signal transmission testing.

4.5 System Performance and Design Specifications

The following baseline specifications define the expected operational and design envelope of the communication subsystem. These parameters establish the reference conditions under which the system shall be designed and evaluated.

Teams may propose justified deviations from these values, provided that technical rationale and performance impact are clearly documented.

Parameter	Target Specification	Notes
Orbit Model	LEO (500–700 km equivalent)	Simulated orbital environment
Frequency Band	UHF / S-band	Selection shall be technically justified
Communication Protocol	AX.25 or CCSDS-inspired framing	CCSDS is strongly encouraged for teams seeking alignment with professional mission standards
Modulation Scheme	Robust constant-envelope or linear schemes (recommended: 2-FSK/GFSK, BPSK, QPSK)	Selection shall be justified based on robustness, hardware constraints, and implementation complexity
Data Rate	32 kbps – 500 kbps (baseline); up to 1 Mbps optional	Higher rates require demonstrated synchronization margin and KPI performance



Parameter	Target Specification	Notes
Doppler Compensation Capability	Must compensate worst-case LEO Doppler shift for the selected carrier frequency	Teams shall calculate and document expected Doppler based on carrier frequency
Payload Size (per packet)	64–256 bytes	Selection shall consider error probability, retransmission cost, and protocol overhead
Forward Error Correction	Mandatory	Reed-Solomon, Convolutional, LDPC, or equivalent with justification
Retransmission Mechanism	Mandatory	Stop-and-Wait or Selective Repeat with defined timers and retry limits
Command Authentication	Mandatory	Lightweight method with replay protection (counter or timestamp)
Simulated Pass Duration	5–10 minutes	Used for throughput and reliability evaluation
Space Segment Power Budget	≤ 5 W	Must reflect realistic CubeSat resource constraints

5. Documentation Package

Each team shall submit a complete and structured engineering documentation package describing the design, implementation, and validation of the communication subsystem. The documentation shall demonstrate technical understanding, design justification, and compliance with the defined functional and performance requirements. The submission shall include the following packages:

5.1 System Design Report

The System Design Report shall describe the overall architecture and engineering rationale of the communication subsystem. It must clearly explain design decisions and demonstrate traceability between requirements and implementation. At a minimum, the report shall include:

- A complete system architecture block diagram
- Link budget analysis
- Modulation and coding selection with technical justification
- Communication protocol selection (e.g., AX.25 or CCSDS-based framing) with rationale
- Trade-off analysis addressing performance, complexity, and resource constraints



5.2 Implementation Description

The implementation description shall document how the proposed architecture was realized in hardware and software. It must provide sufficient technical detail to allow evaluation of system completeness and design maturity. The document shall include:

- Hardware architecture and component description
- Software architecture and module structure
- State machine definitions for communication processes
- Resource usage analysis, including processor utilization, memory footprint, and estimated energy consumption

5.3 Verification Report

The Verification Report shall present the methodology and results of system validation activities. It must demonstrate that the communication subsystem meets the defined functional and performance requirements. The report shall include:

- Description of the test methodology and test setup
- Channel model and impairment configuration used during testing
- Closed-loop operational validation results
- Measured performance metrics aligned with defined KPIs

All test results shall be presented in a structured and quantitative manner.

5.4 User & Operations Manual

The User and Operations Manual shall describe how the communication subsystem is operated from the ground segment and how data is structured and interpreted. This document shall include:

- Ground station operating procedures
- Telecommand format specification
- Telemetry format documentation
- Operational sequence description for a simulated satellite pass

The manual shall be written clearly enough to allow an independent operator to conduct system testing and demonstration.

6. Key Performance Indicators

System performance shall be quantitatively evaluated using defined KPIs. These metrics provide objective measures of communication reliability, efficiency, robustness, and resource utilization. All reported values shall be derived from documented test conditions



and aligned with the evaluation procedures defined in Section 7. The following KPIs shall be assessed:

KPI	Description
Telecommand Reliability	Probability of successful command delivery
Acknowledgment Confirmation Time	Time to verified acknowledgment
Packet Error Rate	Ratio of incorrectly received packets under defined signal-to-noise conditions
Doppler Robustness	Measured performance degradation under defined Doppler shift and Doppler rate conditions
Time to First Lock	Time required to achieve initial synchronization and successful frame decoding
Net Throughput per Pass	Total useful payload data successfully delivered during a simulated pass
Energy per Delivered Bit	Estimated energy consumption per successfully delivered payload bit
Onboard Resource Utilization	Processor load, memory footprint, and estimated power consumption of the space segment

All KPI results shall be presented quantitatively and supported by measurement methodology and test configuration details.

7. Evaluation and Test Plan

The communication subsystem will be evaluated through a structured, multi-stage assessment process designed to verify engineering completeness, technical performance, and operational functionality. Evaluation will be conducted in three phases, with weighted scoring assigned to each stage.

7.1 Phase 1 – Design Review (30%)

The design review phase assesses the completeness and technical maturity of the proposed system architecture prior to full experimental validation. Teams shall present their overall design approach, implementation strategy, and engineering trade-offs. Evaluation during this phase will consider:

- System architecture completeness
- Consistency between requirements and implementation
- Technical justification of design choices
- Quality and clarity of submitted documentation
- Soundness of link budget and performance analysis

This phase focuses on engineering rigor and structured system design.



7.2 Phase 2 – Hardware-in-the-Loop (HIL) Simulation (35%)

In this phase, the communication subsystem will be evaluated under controlled test conditions using SDR-based channel emulation or equivalent testing infrastructure. The objective is to measure system performance under simulated operational impairments. The test environment may include:

- Doppler shift and Doppler rate simulation
- Additive noise
- Signal fading
- Burst interference scenarios

Performance will be quantitatively assessed using measured metrics such as:

- Packet error rate (PER)
- Frame error rate (FER)
- Time to acquisition and synchronization
- Retransmission efficiency and acknowledgment latency

All measurements shall be documented and traceable to defined Key Performance Indicators (KPIs).

7.3 Phase 3 – Closed-Loop Demonstration (35%)

The final phase evaluates complete end-to-end system integration through a closed-loop operational sequence simulating a satellite pass.

Teams shall demonstrate the following sequence:

- Transmission of a telecommand from the ground segment
- Successful reception and acknowledgment by the space segment
- Execution of a predefined spacecraft state change
- Downlink of telemetry confirming the executed action

Successful completion of the full sequence is required to demonstrate system-level functionality. Failure to complete any stage of the operational chain will result in incomplete validation of the subsystem.